RAMCO INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

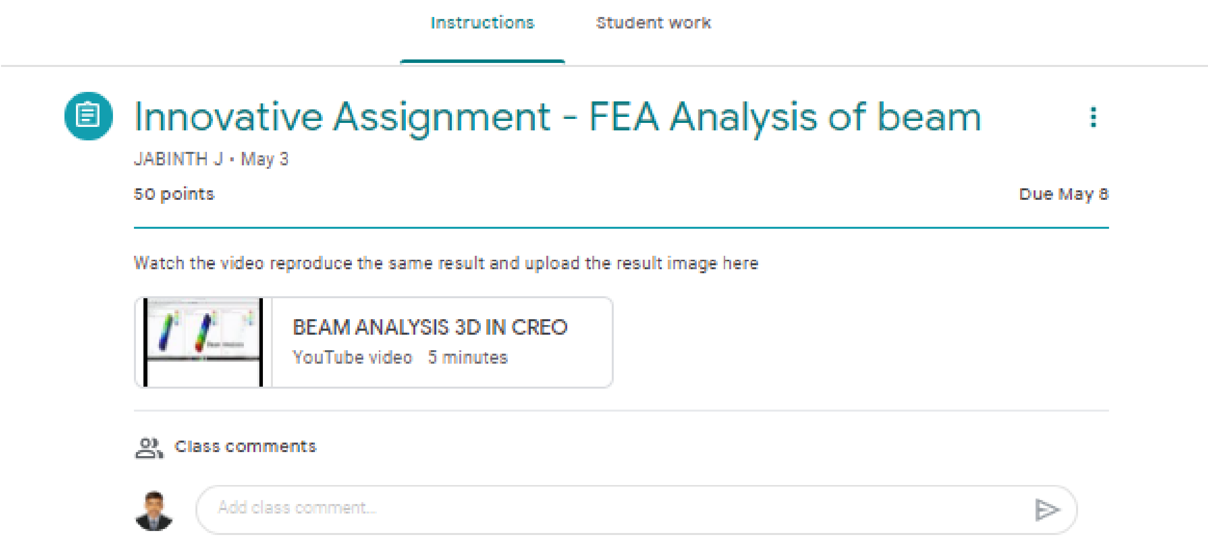
Academic Year 2020 – 2021 (Even Semester)

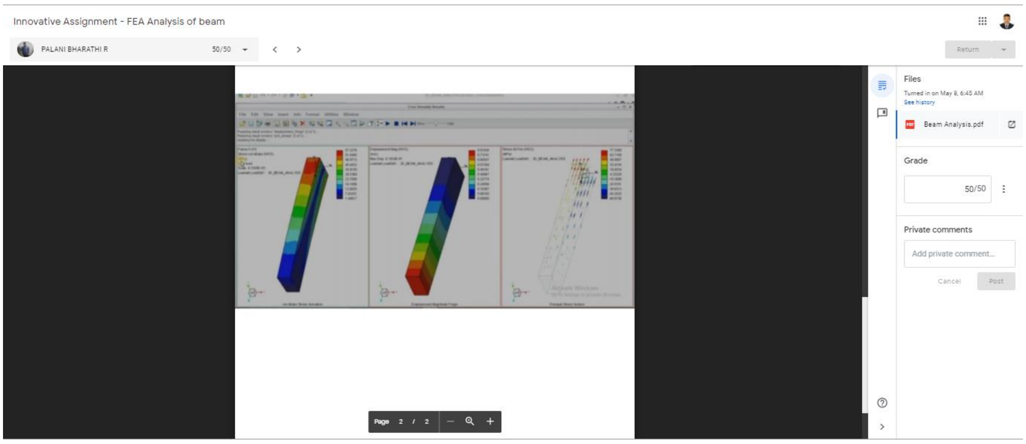
Innovative Practice(s) T2P Description (Beam FEA using CREO)

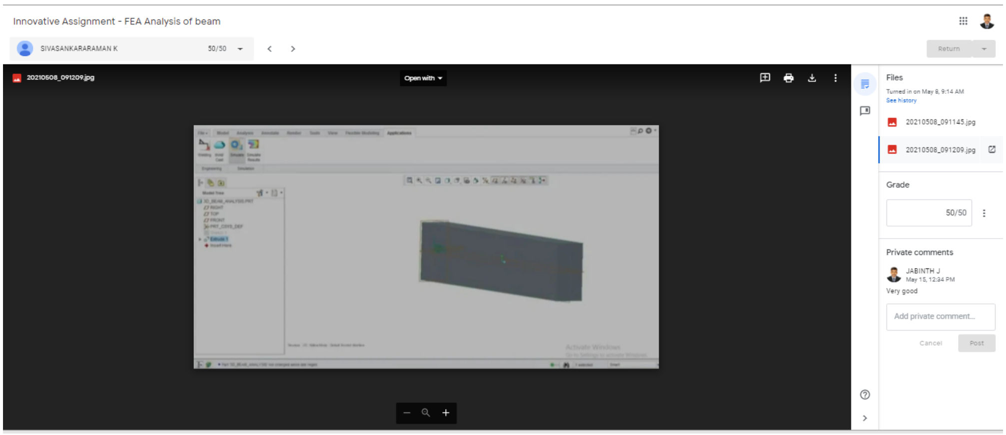
Course code and Title	:	ME8692 – Finite Element Analysis
Name of the Instructor(s)	:	Mr. J. Jabinth, AP / Mechanical
Date & Time	:	03.05.2021
Semester and Branch	:	VI Semester –Mechanical Engineering

Finite Element Analysis of Cantilever Beam Using CREO

CREO is 3D modeling parametric based tool, which can also be used for FEA analysis. Students have learned the basic concepts and the method to calculate displacement and stresses in a beam by hand calculation. In order to give an experience using FEA packages, they are asked to watch a video of CREO analysis and reproduce the same result and also try for different cross-section. The screenshots are shown below,







OUTCOME:

- CO2 – Apply finite element formulations to solve two dimensional scalar Problems
- CO5 – Apply finite element method to solve problems on Iso parametric element and dynamic Problems.

This activity helps in attaining the following CO – PO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO2	3	3	3	1	3				3	3		1	3	2	1	2
CO5	3	3	3	1	3				3	3		1	3	2	1	2